

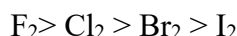
HALOGENS

Reactivity

The reactivity of halogens is directly related to their ability to gain an electron and form halide ion. As you go down the group, the electronegativity of halogens decreases. This results in a decreasing down reactivity.

Due to their strong tendency to gain electrons halogens have strong oxidizing power and this power decreases down the group.

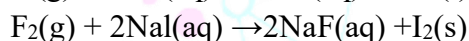
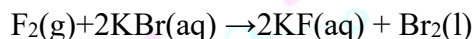
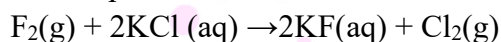
Thus, the order of decreasing oxidizing power is:



Relative strength as oxidizing agent

Relative strength as oxidizing agent, it is possible for a free halogen to oxidize or displace the ion of halogen next to it in the group from their aqueous solutions.

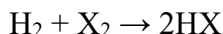
- This means F_2 can oxidize and displace all the halide ions to free halogen. For example,



- Similarly, Cl_2 can oxidize Br and I ions.
- But I_2 cannot oxidize any halide ion.

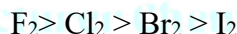
Reactions of halogens with hydrogen

Halogens react with hydrogen to form hydrogen halides.



Where $X = F_2, Cl_2, Br_2, I_2$

The relative reactivity of halogens in these reactions decreases from fluorine to iodine



- ✓ Fluorine combines explosively with hydrogen even in the cold and darkness to form hydrogen fluoride gas.
- ✓ Chlorine and hydrogen explode when exposed to sunlight or flame to form hydrogen chloride gas.
- ✓ Bromine vapour and hydrogen together produce a small explosion when ignited. Hydrogen bromide gas is produced.
- ✓ Iodine and hydrogen combine only partially even with continued heating. An equilibrium is formed between hydrogen and iodine and hydrogen iodide gas.

Thermal stability of Hydrogen halides

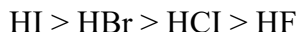
A larger electronegativity difference results in a stronger bond. As we move from HF to HI, the electronegativity difference between the hydrogen and halogen atoms decreases, resulting in weaker bonds.

So, the relative thermal stability of hydrogen halides gradually decreases from HF to HI. The energy needed to break H-X decreases in the following orders

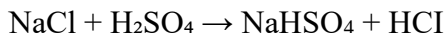
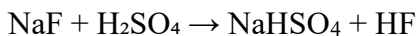


Acidic Strength of Halogen acids

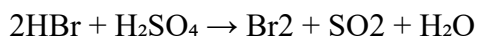
The acidic strength of halogen acids decreases in the following order.

**Reactivity of Halides Ions****1. With H₂SO₄:**

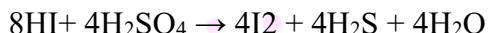
- Fluoride, F⁻ and chloride, Cl⁻ ions are weak reducing agents so they can not reduce concentrated sulphuric acid.



- Bromide ions have enough strong reducing and can reduce concentrated sulphuric acid to sulphur dioxide.



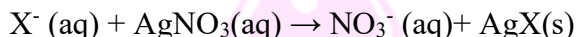
- Iodide ions are even stronger reducing agent and can reduce concentrated sulphuric acid to hydrogen sulphide.

**2. Reaction with silver nitrate**

This reaction is to identify halide ions and is called the **silver nitrate test**.

When the aqueous solution of silver nitrate is treated with aqueous solution of a halide salt. Colored precipitates are formed.

- Chlorides form white ppt which remain un-dissolved on adding aqueous ammonia.
- A bromide forms cream or off-white ppt, which partially dissolves in aqueous ammonia.
- Whereas, an iodide forms yellow ppt, which remains un-dissolved in aqueous ammonia.

**3. Reaction with sodium hydroxide**

Chlorine reacts with NaOH in two ways.

When chlorine gas is passed through cold and dilute aqueous solution of NaOH, it forms sodium hypochlorite (NaOCl).



When chlorine is passed through hot and concentrated solution of NaOH, it forms Sodium chlorate (NaClO₃)

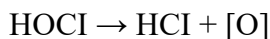
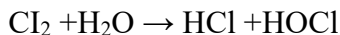


(self-oxidation-reduction of Cl or disproportion reaction)

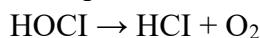
4. Reaction of chlorine with water

Chlorine dissolves in water producing hydrochloric acid (HCl) and hypochlorous acid (HOCl), which is a very reactive species.

It decomposes to produce atomic oxygen which can kill bacteria present in water. For this reason, Cl₂ is used in the purification of bacteria. This process is called chlorination.



In sunlight or on standing, HOCl decomposes to produce O₂.



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